D02. Warehouse Process Flow and Business Rules

Process Overview

This project simulates a simplified warehouse flow within the FlowCore environment. Incoming goods arrive at an inbound dock and are presented in the system as inbound pallets or equivalent units. Each pallet is assigned to a valid warehouse location and according to predefined configuration and simple rules-based logic. During this process, status changes and operational event records are generated so that the progress of each pallet can be traced and validated. The flow ends only when all successfully processed pallets have reached the expected stored state and the environment still behaves correctly in the case that the current configuration or logic changes.

Main Process Flow

- **Incoming goods arrive at the inbound dock**
- **The incoming goods are represented in the system as inbound pallets**
- **The system evaluates the available warehouse location based on predefined configuration**
- **Each pallet is assigned to one valid warehouse location**
- **Status changes and event records are created to capture each pallet through the inbound flow.**
- **Validation checks confirm that the final environment state matches the expected result.**

Business rules

- **Every inbound pallet must belong to a valid inbound reference.**
- **Every pallet must be assigned to one valid warehouse location.**
- **A pallet must not be assigned to an invalid or unavailable location.**
- **A pallet must not remain unassigned after successful processing.**
- **Each important process step must generate a status or event record.**
- **Only valid forward status transitions are allowed.**

- **A pallet cannot reach the final stored state without first being received and assigned.**
- **Location assignment must follow the current configuration and logic.**
- **The final system state must be consistent with the expected inbound process result.**
- **Change in the system logic must not result in unexpected invalid behavior.**